

Geometric Numerical Integrators

The Störmer-Verlet Method and Its Properties

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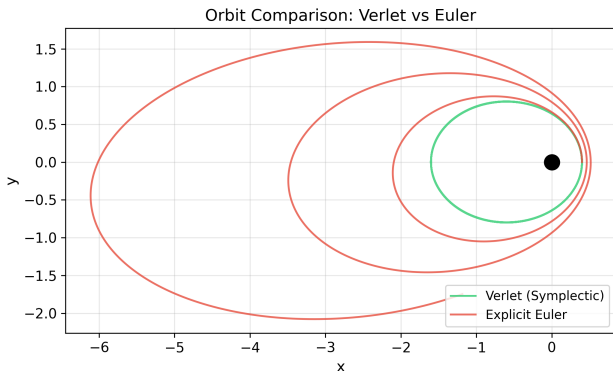
Why Do Geometric Integrators Matter?

Problems with classical methods:

- Work well in the short term
- Long term: **energy drift**
- Trajectory distortion
- Physically incorrect results

Application areas:

- Celestial mechanics
- Molecular dynamics (10^6 – 10^9 steps!)
- Vibrational mechanical models



Hamiltonian Systems

Definition

Hamiltonian systems are described by the differential equations:

$$\dot{p} = -\nabla_q H(p, q), \quad \dot{q} = \nabla_p H(p, q)$$

where $H(p, q)$ is the *Hamiltonian function* (energy).

Separable Hamiltonian

Common special case in physical systems:

$$H(p, q) = \underbrace{\frac{1}{2} p^T M^{-1} p}_{\text{kinetic energy}} + \underbrace{U(q)}_{\text{potential energy}}$$

Second-order form: $\ddot{q} = f(q)$, where $f(q) = -M^{-1} \nabla U(q)$.

Geometric Properties of Hamiltonian Systems

- ① **Symplecticity:** The flow φ_t preserves the symplectic structure:

$$\varphi'_t(p, q)^T J \varphi'_t(p, q) = J$$

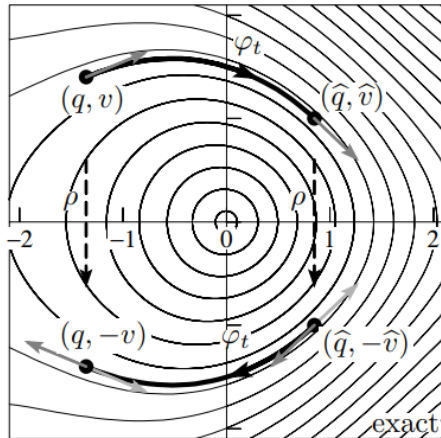
where $J = \begin{pmatrix} 0 & I \\ -I & 0 \end{pmatrix}$

$\Rightarrow$ Area preservation (Liouville's theorem)

- ② **Energy conservation:** $H(p(t), q(t)) = \text{constant}$
 $\Rightarrow$ The Hamiltonian is a *first integral*.

- ③ **Reversibility:** Reversing time ($t \rightarrow -t$, $p \rightarrow -p$) retraces the original trajectory

Goal: Numerical methods that preserve these properties!



Derivation of the Störmer-Verlet Method

Starting point: $q'' = f(q)$ second-order differential equation

Discretization: $\frac{q_{n+1} - 2q_n + q_{n-1}}{h^2} = f(q_n)$

Introducing auxiliary variables ($v = q'$):

$$v_{n+\frac{1}{2}} = \frac{q_{n+1} - q_n}{h}, \quad v_n = \frac{q_{n+1} - q_{n-1}}{2h}, \quad q_{n+\frac{1}{2}} = \frac{q_n + q_{n-1}}{2}$$

Störmer-Verlet (A)

$$(q_n, v_n) \rightarrow (q_{n+1}, v_{n+1})$$

$$v_{n+\frac{1}{2}} = v_n + \frac{h}{2} f(q_n)$$

$$q_{n+1} = q_n + h v_{n+\frac{1}{2}}$$

$$v_{n+1} = v_{n+\frac{1}{2}} + \frac{h}{2} f(q_{n+1})$$

Störmer-Verlet (B)

$$(q_{n-\frac{1}{2}}, v_{n-\frac{1}{2}}) \rightarrow (q_{n+\frac{1}{2}}, v_{n+\frac{1}{2}})$$

$$q_n = q_{n-\frac{1}{2}} + \frac{h}{2} v_{n-\frac{1}{2}}$$

$$v_{n+\frac{1}{2}} = v_{n-\frac{1}{2}} + h f(q_n)$$

$$q_{n+\frac{1}{2}} = q_n + \frac{h}{2} v_{n+\frac{1}{2}}$$

Properties: Second-order, explicit for $f(q)$ -type equations

Sub-operators: SE1 and SE2

Split the methods into smaller sub-operators!

SE1 method:

$$(q_n, v_n) \rightarrow (q_{n+\frac{1}{2}}, v_{n+\frac{1}{2}})$$

$$v_{n+\frac{1}{2}} = v_n + \frac{h}{2} f(q_n)$$

$$q_{n+\frac{1}{2}} = q_n + \frac{h}{2} v_{n+\frac{1}{2}}$$

SE2 method:

$$(q_{n+\frac{1}{2}}, v_{n+\frac{1}{2}}) \rightarrow (q_{n+1}, v_{n+1})$$

$$q_{n+1} = q_{n+\frac{1}{2}} + \frac{h}{2} v_{n+\frac{1}{2}}$$

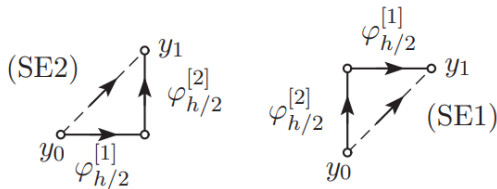
$$v_{n+1} = v_{n+\frac{1}{2}} + \frac{h}{2} f(q_{n+1})$$

Composition

The Störmer-Verlet methods:

$$(A) = (\text{SE2}) \circ (\text{SE1})$$

$$(B) = (\text{SE1}) \circ (\text{SE2})$$



Geometric Interpretation of the Method

Splitting method:

Split the vector field:

$$(v, f(q)) = (v, 0) + (0, f(q))$$

Two sub-operators:

$$\varphi_t^{[1]} : \begin{cases} q_1 = q_0 + tv_0 \\ v_1 = v_0 \end{cases} \quad (\text{drift})$$

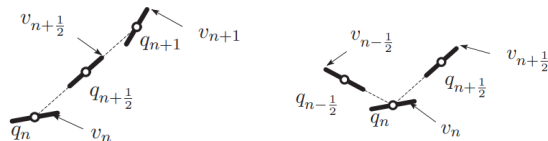
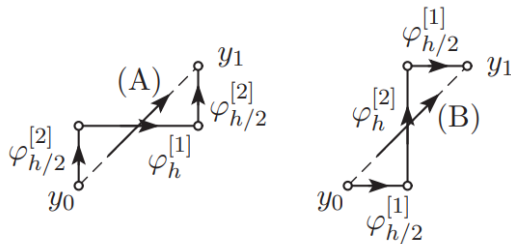
$$\varphi_t^{[2]} : \begin{cases} q_1 = q_0 \\ v_1 = v_0 + tf(q_0) \end{cases} \quad (\text{kick})$$

Störmer-Verlet (A):

$$\Phi_h^{(A)} = \varphi_{h/2}^{[2]} \circ \varphi_h^{[1]} \circ \varphi_{h/2}^{[2]}$$

Störmer-Verlet (B):

$$\Phi_h^{(B)} = \varphi_{h/2}^{[1]} \circ \varphi_h^{[2]} \circ \varphi_{h/2}^{[1]}$$



Generalization to Arbitrary Systems

Question: Can we use Störmer-Verlet for general first-order systems?

Original: $q'' = f(q)$

$$\begin{cases} q' = v \\ v' = f(q) \end{cases}$$

Methods are explicit:

- Direct computation
- Low computational cost
- Ideal for this form!

General: arbitrary system

$$\begin{cases} q' = g(q, v) \\ v' = f(q, v) \end{cases}$$

Methods become implicit:

$$\begin{aligned} v_{n+\frac{1}{2}} &= v_n + \frac{h}{2} f(q_n, v_{n+\frac{1}{2}}) \\ q_{n+1} &= q_{n+\frac{1}{2}} + \frac{h}{2} g(q_{n+1}, v_{n+\frac{1}{2}}) \end{aligned}$$

Conclusion

Störmer-Verlet is best suited for $q'' = f(q)$ type equations!

For general systems, the implicit solves become computationally expensive.

What is Symplecticity?

Definition

A numerical integrator Φ_h is *symplectic* if

$$\Phi'_h(p, q)^T J \Phi'_h(p, q) = J$$

at every point (p, q) .

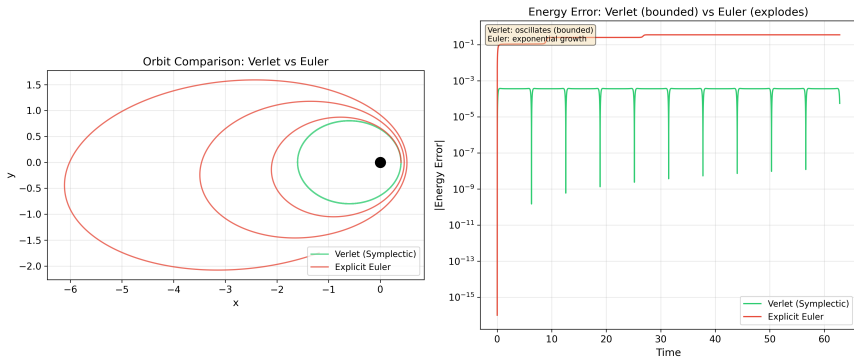
Consequences:

- Preserves phase space volume (Liouville's theorem)
- **Bounded energy error** in the long term (no drift!)
- Physically faithful simulation

Theorem: Symplecticity of Störmer-Verlet

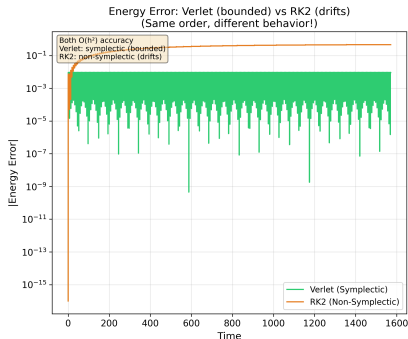
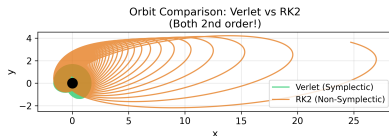
The (SE1) and (SE2) sub-operators are symplectic, therefore their composition, the Störmer-Verlet (A) and (B) methods, are also symplectic.

Störmer-Verlet vs. Explicit Euler



- **Left:** Trajectories over 10 orbits ($h = 0.01$)
- **Right:** Energy error vs. time
- **Euler:** exponential energy drift, spiral trajectory
- **Verlet:** bounded oscillating error, stable ellipse

Störmer-Verlet vs. RK2: The Key Comparison

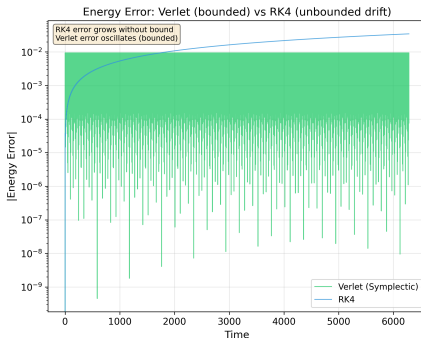
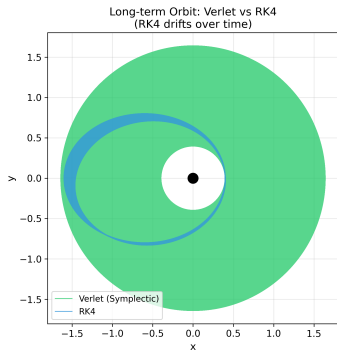


Both methods are second-order ($O(h^2)$)!

- **Verlet (symplectic):** Perfect ellipse after 250 orbits
- **RK2 (non-symplectic):** Trajectory drifts, energy grows

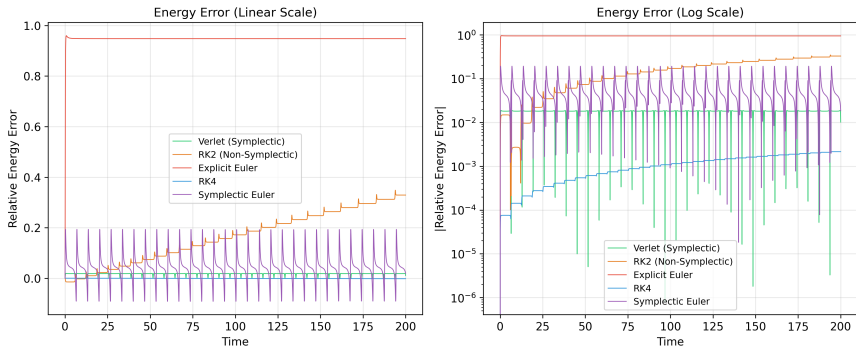
Symplecticity is more important than accuracy order!

Störmer-Verlet vs. RK4: Long-Term Behavior



- **RK4**: 4th-order accuracy, but **not symplectic**
- Short term (less than 100 orbits): RK4 is more accurate
- Long term (more than 100 orbits): **Verlet wins!**
- RK4 energy drifts, Verlet remains bounded

Energy Error Comparison: All Methods



Dichotomy:

- **Symplectic methods:** bounded oscillating error
- **Non-symplectic methods:** energy drift

Liouville's Theorem and Area Preservation

Liouville's Theorem

The flow of Hamiltonian systems preserves phase space volume:

$$\det \varphi'_t(p, q) = 1 \quad \forall t$$

For numerical methods: A method preserves area if

$$\det \frac{\partial}{\partial(p_0, q_0)} \Phi_h = 1$$

Example - Pendulum motion: $p' = q$, $q' = -\sin p$

Explicit Euler:

$$\det \Phi_h^{EE} = 1 + h^2 \cos p_0 \neq 1$$

Implicit Euler:

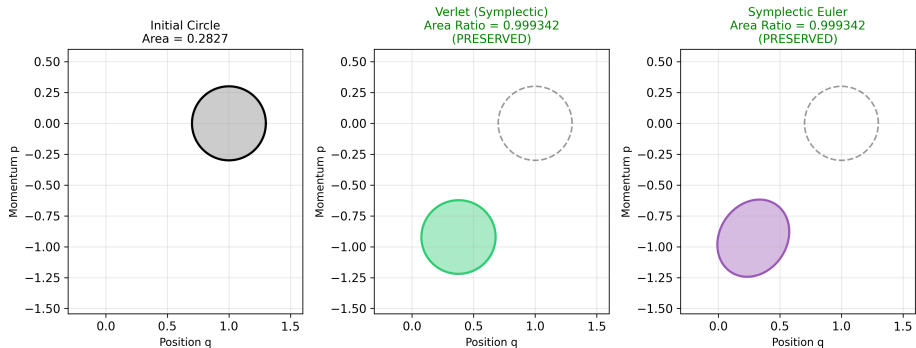
$$\det \Phi_h^{IE} = \frac{1}{1 + h^2 \cos p_1} \neq 1$$

SV (SE1, SE2):

$$\det \Phi_h^{SE1} = \det \Phi_h^{SE2} = 1 \quad \checkmark$$

Visual Demonstration: Symplectic Methods

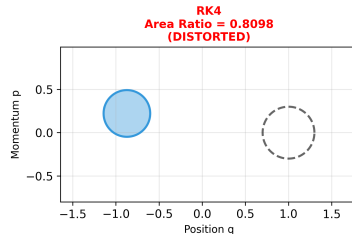
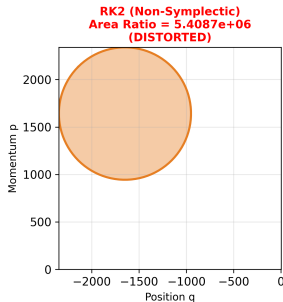
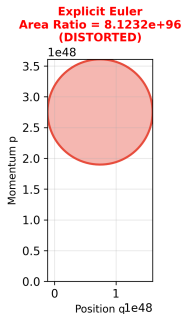
Symplectic Methods: Area is Preserved
(Circle rotates/transforms but maintains area)



- Circle of initial conditions in phase space, 100 steps ($h = 0.2$)
- **Verlet & Sympl. Euler:** Area ratio ≈ 1.000 (within 0.1%)
- The circle rotates and deforms, but **area is preserved**

Visual Demonstration: Non-Symplectic Methods

Non-Symplectic Methods: Area is NOT Preserved
(Circle expands/contracts, violating Liouville theorem)



- Same test, 1000 steps, $h = 0.5$

- **Euler:** Drastic expansion

- **RK2:** Significant distortion

- **RK4:** Measurable distortion

Symmetry and Reversibility

Time Reversibility

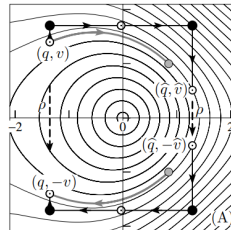
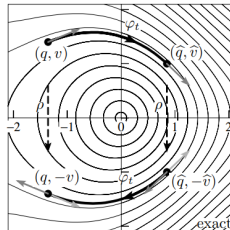
Hamiltonian systems are reversible: if we reverse time and momenta ($t \rightarrow -t$, $p \rightarrow -p$), we retrace the original trajectory.

Störmer-Verlet is symmetric:

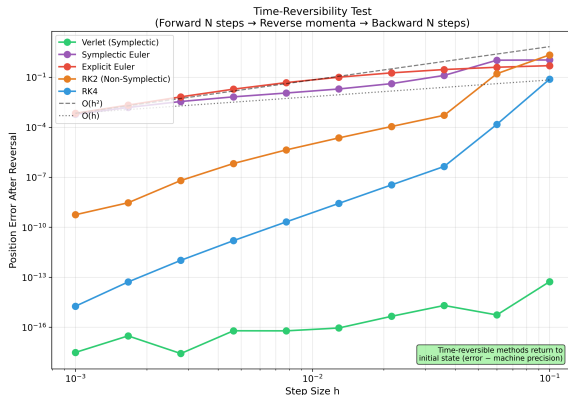
$$\begin{aligned}\text{Forward: } v_{n+\frac{1}{2}} &= v_n + \frac{h}{2}f(q_n) \\ q_{n+1} &= q_n + hv_{n+\frac{1}{2}} \\ v_{n+1} &= v_{n+\frac{1}{2}} + \frac{h}{2}f(q_{n+1})\end{aligned}$$

If we reverse: $h \rightarrow -h$, $v \rightarrow -v$, we **exactly recover** (q_n, v_n) !

Kick-Drift-Kick $\Rightarrow$ symmetry



Reversibility Test



Test: N steps forward $\rightarrow$ momentum reversal $\rightarrow N$ steps backward

- **Verlet:** Error $\approx 10^{-14}$ (machine precision!)
- Euler, RK2, RK4: Errors scale with $O(h^p)$
- Only symmetric methods are perfectly reversible

Two-Particle Lennard-Jones Simulation

Lennard-Jones potential:

Interaction of noble gases, simple molecules

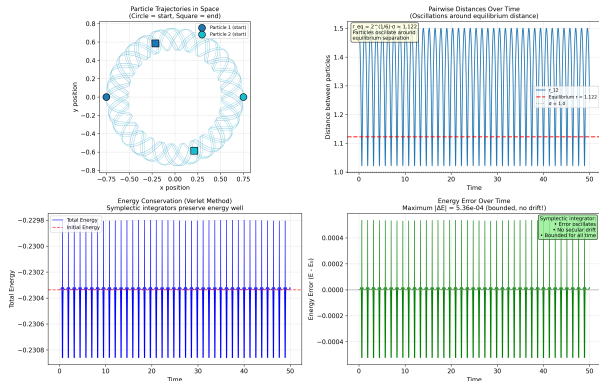
$$V_{LJ}(r) = 4\epsilon \left[\left(\frac{\sigma}{r} \right)^{12} - \left(\frac{\sigma}{r} \right)^6 \right]$$

- r^{-12} : Pauli repulsion
- r^{-6} : van der Waals attraction
- Equilibrium distance:
 $r_{eq} = 2^{1/6}\sigma \approx 1.122\sigma$

Simulation results:

- **Upper left:** Particle trajectories
- **Upper right:** Distance oscillates around r_{eq}
- **Lower left:** Energy converts
- **Lower right:** Relative energy error stable!

Lennard-Jones Molecular Dynamics (2 particles)



Three-Particle Lennard-Jones Simulation

Many-body system:

- 3 pairwise interactions
- Complex dynamics
- **Energy is conserved!**

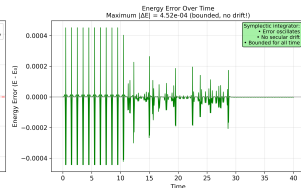
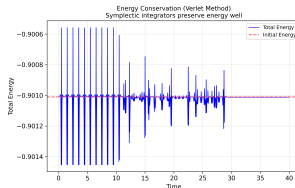
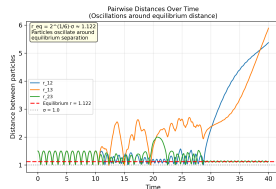
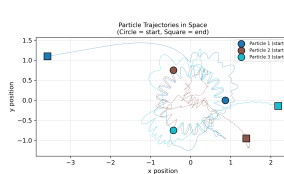
Scalability:

- Larger systems: 10^4 – 10^9 atoms
- Real MD simulations: 10^6 – 10^9 time steps
- **Verlet is the only option** for long-term stability

Industrial applications:

- GROMACS, LAMMPS
- AMBER, NAMD
- Protein dynamics
- Materials design

Lennard-Jones Molecular Dynamics (3 particles)



Comparison of Implemented Methods

Method	Type	Order	Symplectic	Reversible
Störmer-Verlet	Splitting	2	✓	✓
Sympl. Euler	Splitting	1	✓	—
Explicit Euler	Standard	1	—	—
RK2 (Midpoint)	Runge-Kutta	2	—	—
RK4	Runge-Kutta	4	—	—

Key Question

Is it worth using higher-order methods?

The answer: NOT always!

Key Message

For Hamiltonian systems:

Geometric structure $>$ Algebraic accuracy order

Advantages of Störmer-Verlet:

- **Symplectic:** Preserves phase space geometry
- **Bounded energy error:** No drift in the long term
- **Area preservation:** Liouville's theorem satisfied
- **Reversible:** Time-symmetric
- **Efficient:** Only one force evaluation per step

When to use: Conservative systems, long-term simulations

When NOT to use: Dissipative systems, very short integrations

Recommended reading:

- E. Hairer, C. Lubich, G. Wanner: *Geometric Numerical Integration* (2006)
- B. Leimkuhler, S. Reich: *Simulating Hamiltonian Dynamics* (2004)
- J.M. Sanz-Serna, M.P. Calvo: *Numerical Hamiltonian Problems* (1994)

Open-source implementation:

- GitHub repository: Python code + Jupyter notebook
- 14 visualizations + 4 animations
- Full reproducibility
- Detailed documentation (PLOTS_DOCUMENTATION.md, README.md)

Thank you for your attention!

Questions?